

Machine Learning Foundations

Problem types, learning cycles, evaluation, and risks for traditional ML and generative AI

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The business question determines the learning approach

01 Classification

Predict a category, such as fraud versus legitimate activity.

02 Regression

Predict a numeric value, such as demand or delivery time.

03 Clustering

Discover natural groups when labeled outcomes are unavailable.

04 Generation

Produce new text, images, audio, summaries, or code from prompts and context.

Traditional ML specializes; foundation models generalize

Traditional machine learning

Usually trained or tuned for a defined prediction task using structured or specialized datasets.

Foundation models

Pre-trained on broad data and adapted through prompting, retrieval, fine-tuning, or agents.

A model moves through a repeatable learning cycle

01

02

03

04

05

Define

Translate a business need into a measurable ML objective.

Prepare

Collect, clean, transform, label, and split suitable data.

Learn

Train or adapt a model and tune its behavior.

Evaluate

Measure performance, fairness, safety, latency, and cost.

Operate

Deploy, monitor, update, and govern the solution.

Training creates behavior; inference uses it

Training or customization

The system learns patterns or adapts model behavior. This stage can require significant compute, time, and data.

Inference

A trained model receives new input and returns a prediction or generated response under latency and cost constraints.

Evaluation metrics must reflect the cost of error

01 Classification

Precision, recall, F1 score, ROC-AUC, and confusion matrices.

02 Regression

MAE, MSE, RMSE, and R-squared measure numeric error.

03 Generative AI

Quality, relevance, groundedness, safety, human review, and task-specific metrics.

04 Operations

Latency, throughput, availability, drift, and cost reveal production fitness.

Model quality can fail before or after deployment

01 Underfitting

The model is too simple or insufficiently trained to learn useful patterns.

02 Overfitting

The model memorizes training details and performs poorly on unfamiliar data.

03 Bias

Data or model behavior creates systematically unfair or unrepresentative outcomes.

04 Drift

Real-world data or relationships change after deployment, reducing performance.

Machine learning is a managed cycle, not a single algorithm

Start with a measurable business objective.
Match the approach to the data and required output.
Evaluate technical, human, and operational outcomes.
Expect models and data to change over time.

NEXT

**Part 3: The AWS
AI/ML lifecycle**